



A Lyman Continuum–selected search for ionizing emitter galaxies in the HUDF

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REJECTED

The Topsy-Turvy LyC-Leaker Survey

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Background: We **pre-select** galaxies as likely LyC emitters to search efficiently...

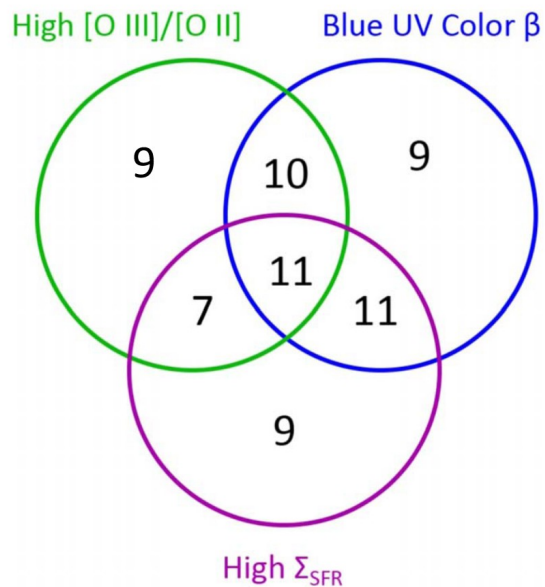


Image: Flury et al., 2022

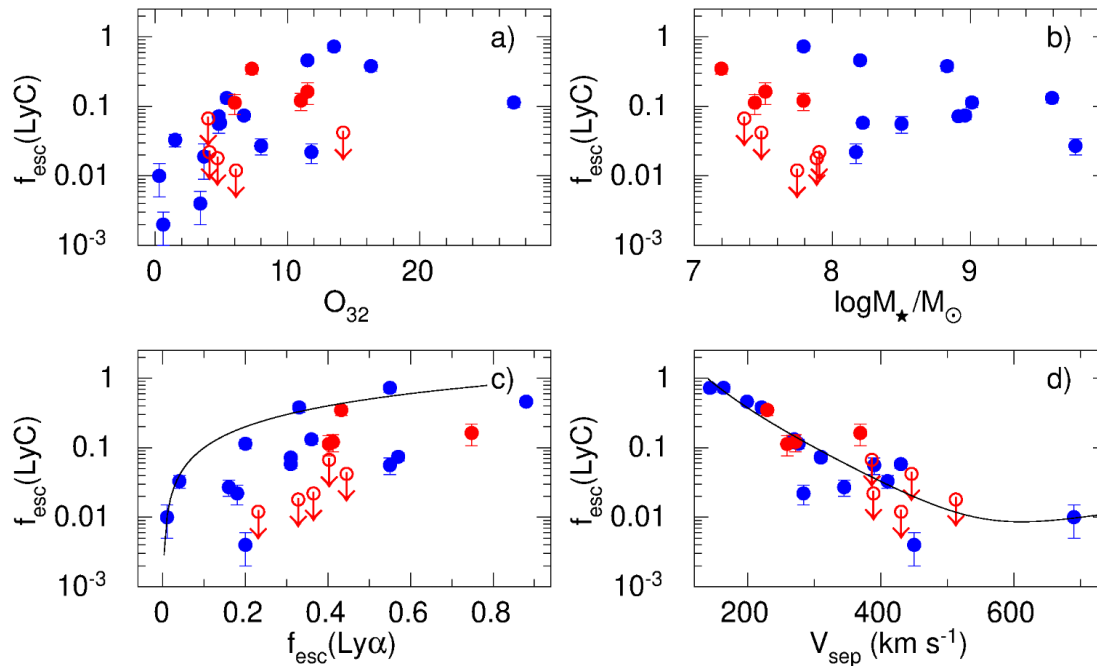
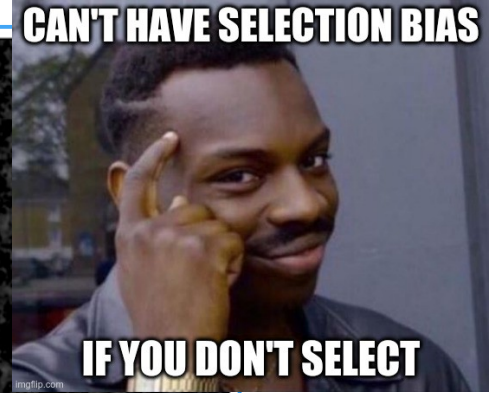
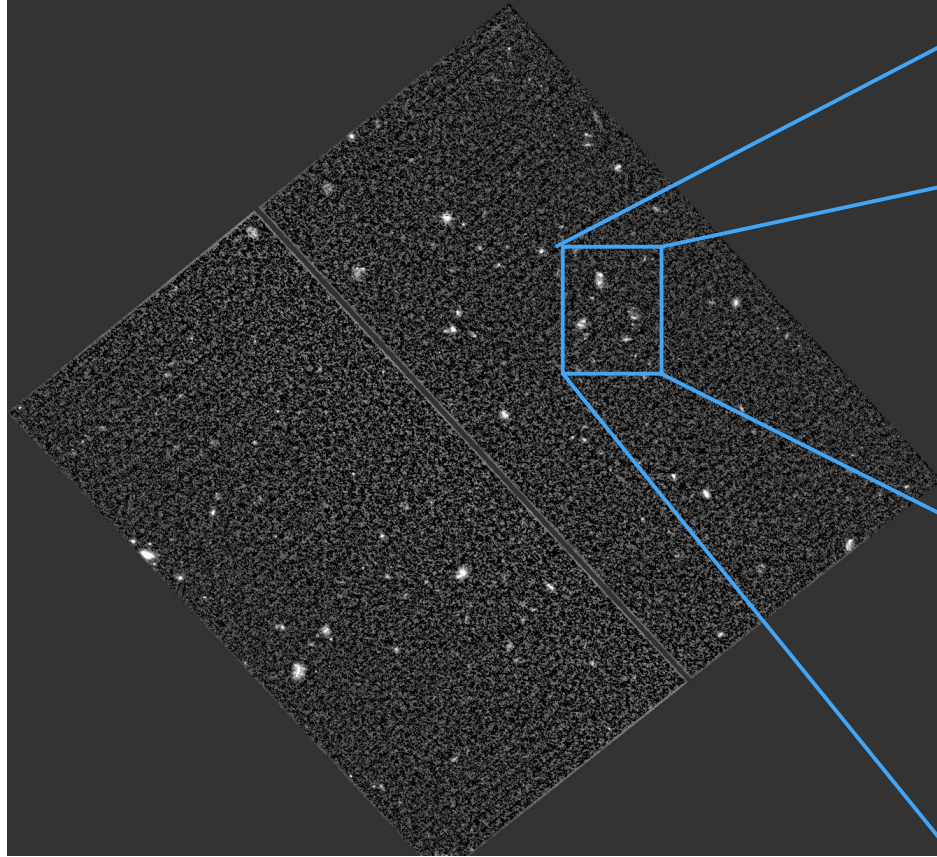


Image: Izotov et al., 2018

But who knows what we **leave out** in that pre-selection?



This work: find **everything** in the UV bands,
then check each source to see if it is escaping LyC



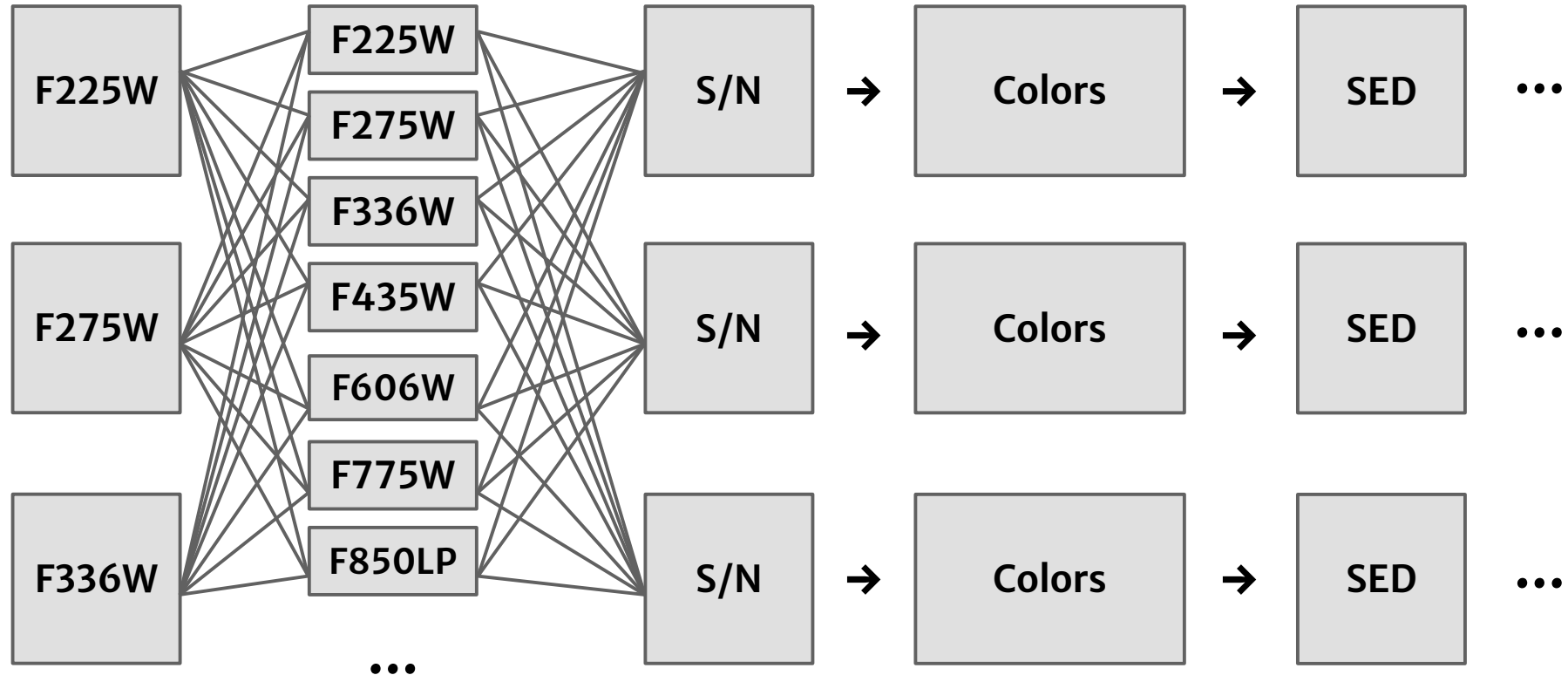
Detect UV sources → define segmentation map → perform photometry → start sorting Müsli



Detection

Photometry

Filtering by various criteria

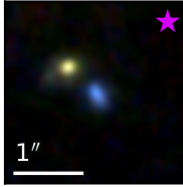


We **compared redshifts** with catalogs in the literature:
Rafelski et al., 2015; Inami et al., 2017

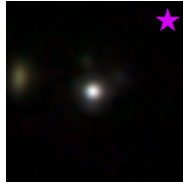


We ended up with **7 candidate leakers**;
4 of which we consider “very convincing”

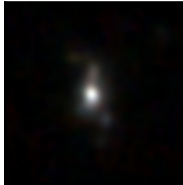
F336W-1041



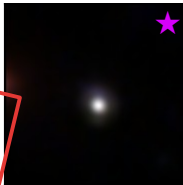
F336W-1013



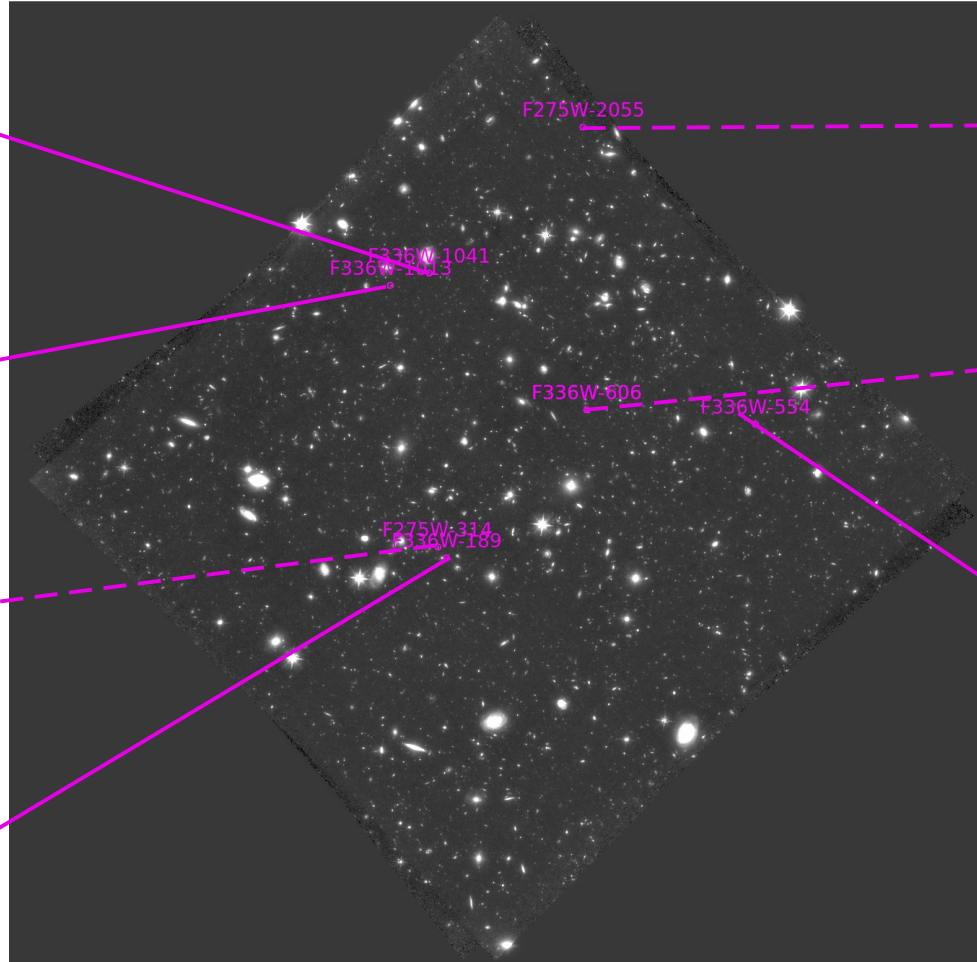
F275W-314



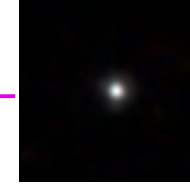
F336W-189



SEE ALSO:
Saxena et al., 2021



F275W-2055



F336W-606

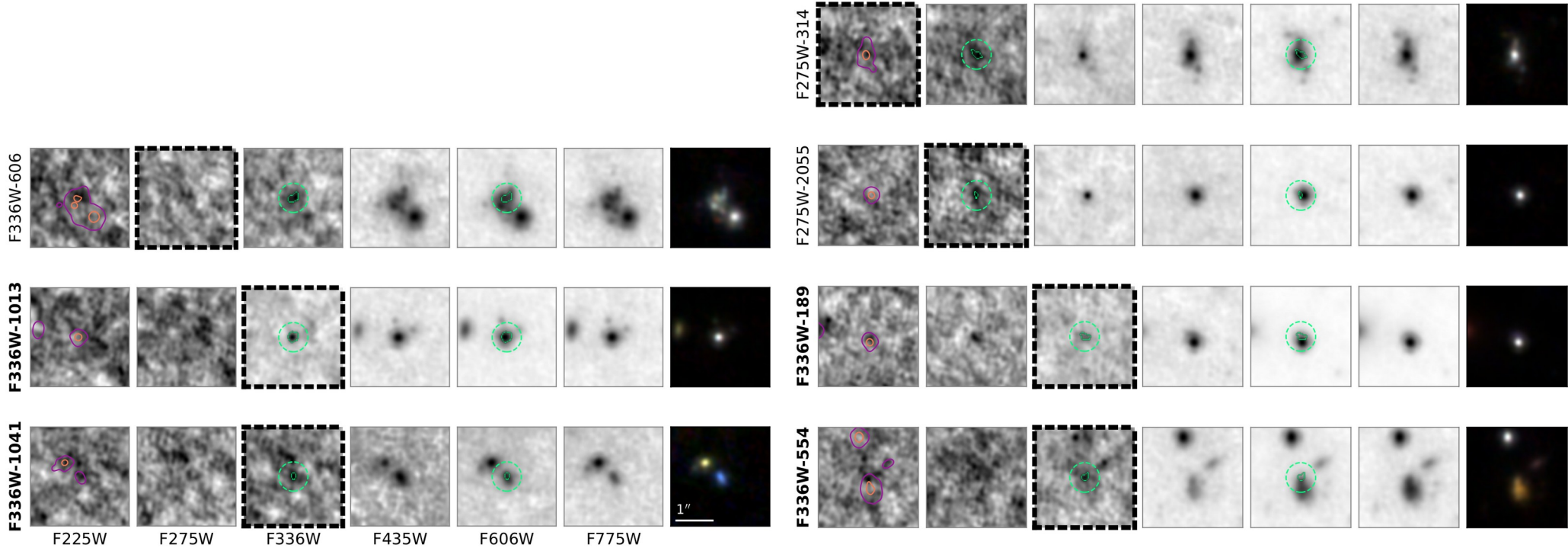


F336W-554



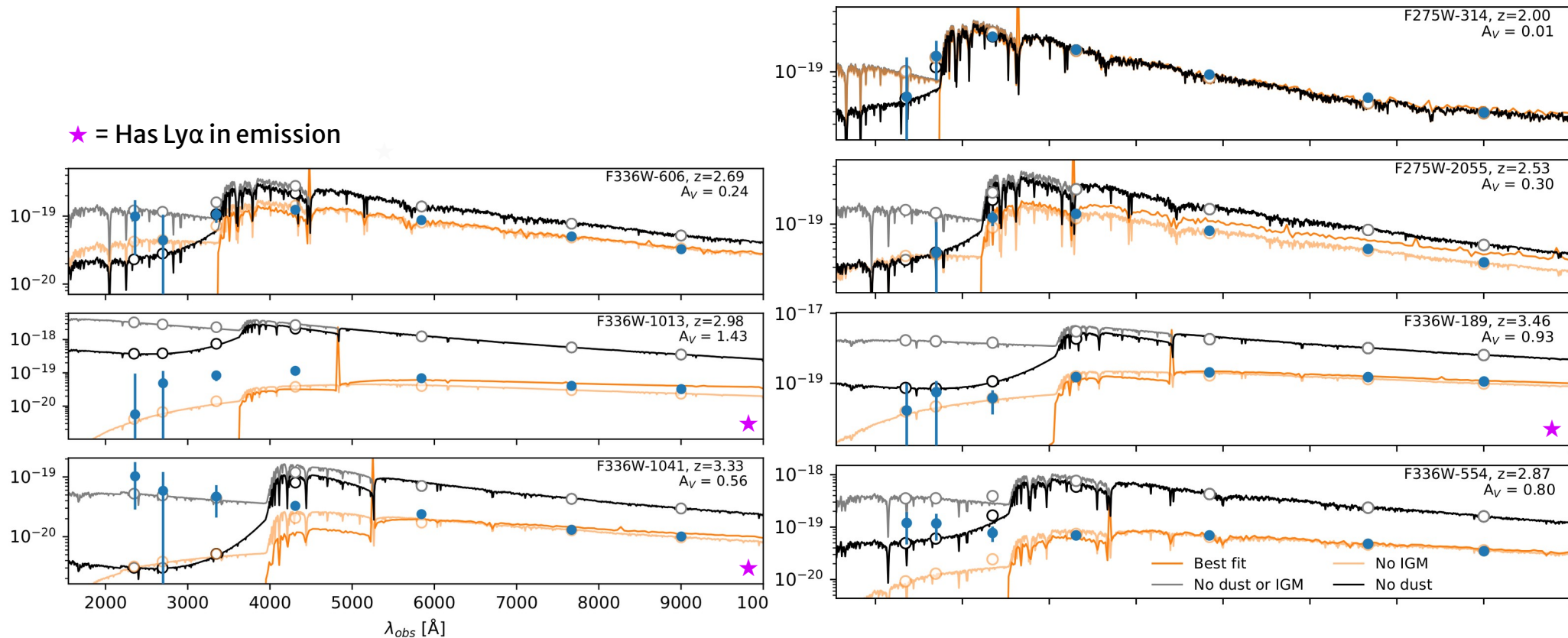
★: Has Ly α in emission

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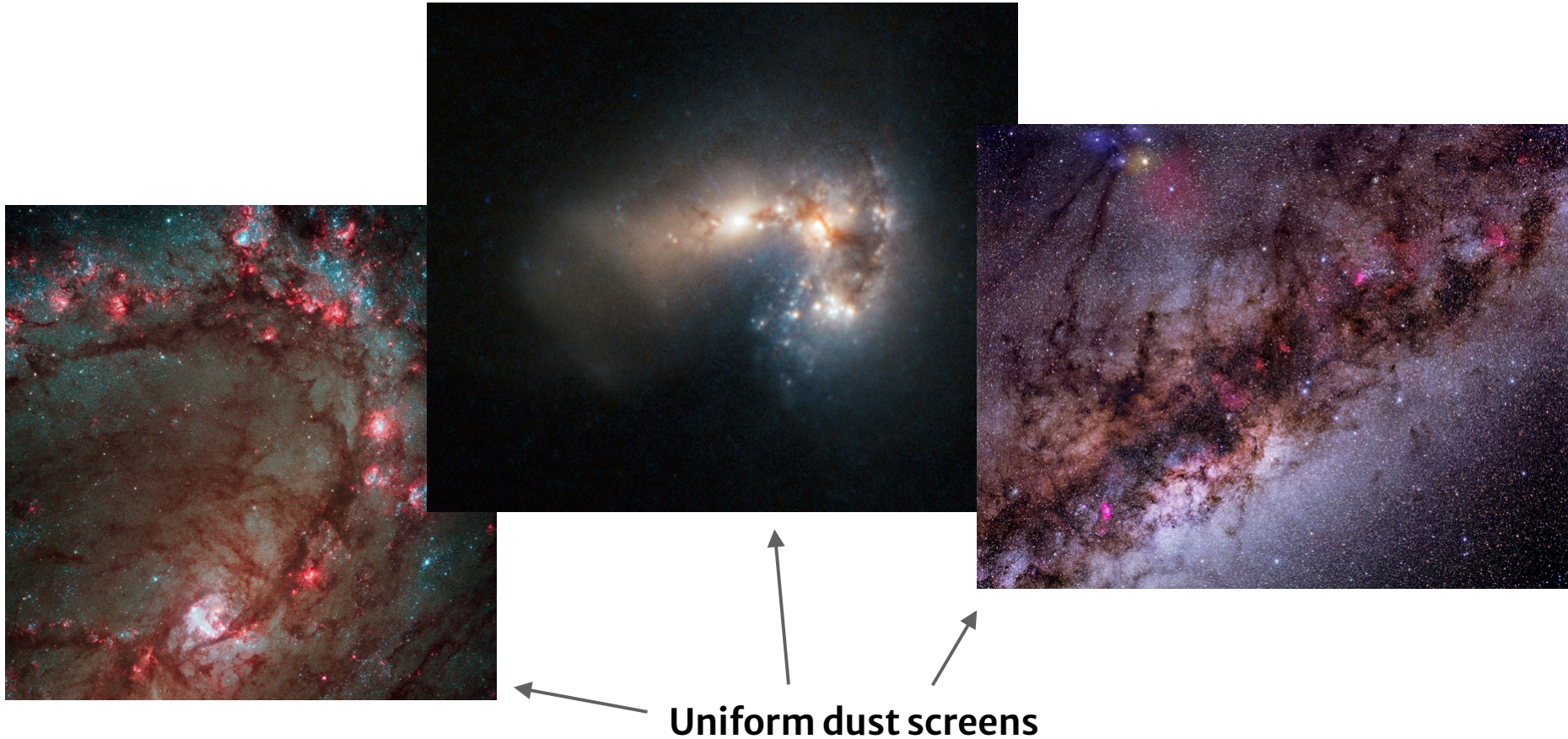
We computed absolute and relative ionizing **escape fractions** from SED models

★ = Has Ly α in emission





Problem: we convert between absolute and relative escape fractions assuming **a uniform dust screen**



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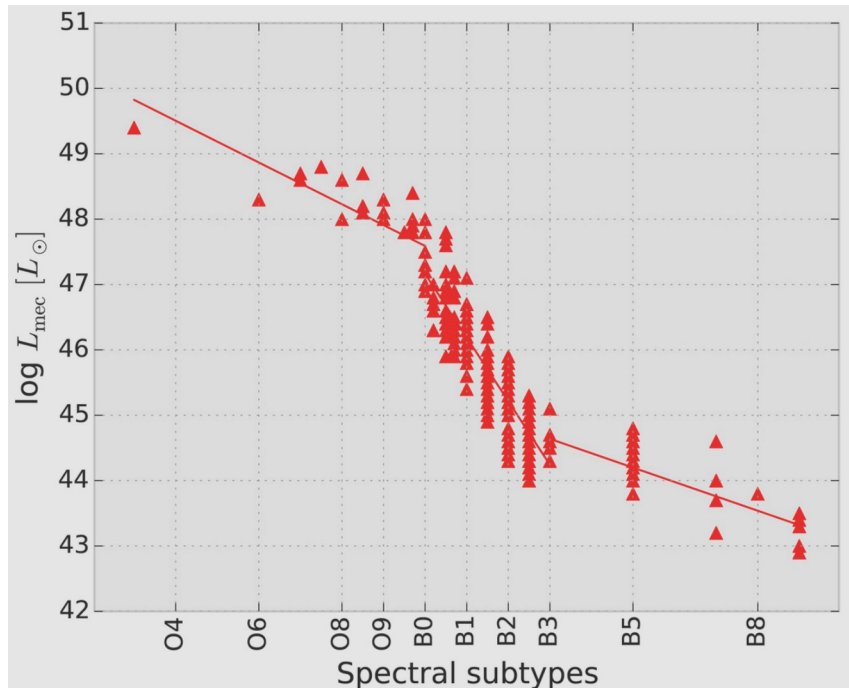


$$f_{\text{esc}}^{\text{rel}} = \frac{(f_{1500}/f_{900})_{\text{Int}}}{(f_{1500}/f_{900})_{\text{Obs}}} T_{\text{IGM}}, \quad (1)$$

$$f_{\text{esc}}^{\text{abs}} = f_{\text{esc}}^{\text{rel}} \times 10^{-0.4 \times A_{1500}}, \quad (2)$$

LyC emission is dominated by so few stars that their experienced τ_{dust} can get **highly stochastic**

The ionizing output of a galaxy can be dominated by a few very bright stars – as few as a handful.

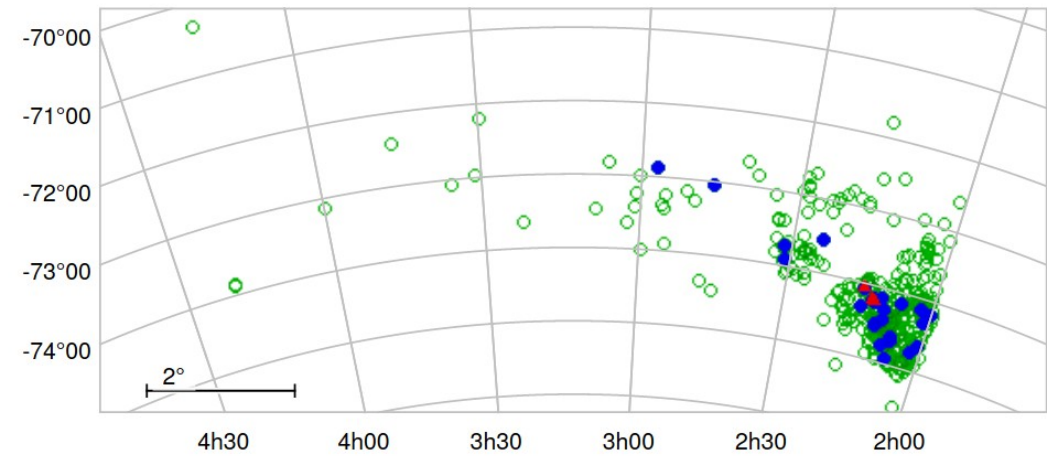


E.g. V. Ramachandran+, 2019

SEE ALSO Marta
Lorenzos talk yesterday!

These stars are not always where the gas and the general population is.

There are O-stars in the Magellanic Bridge!

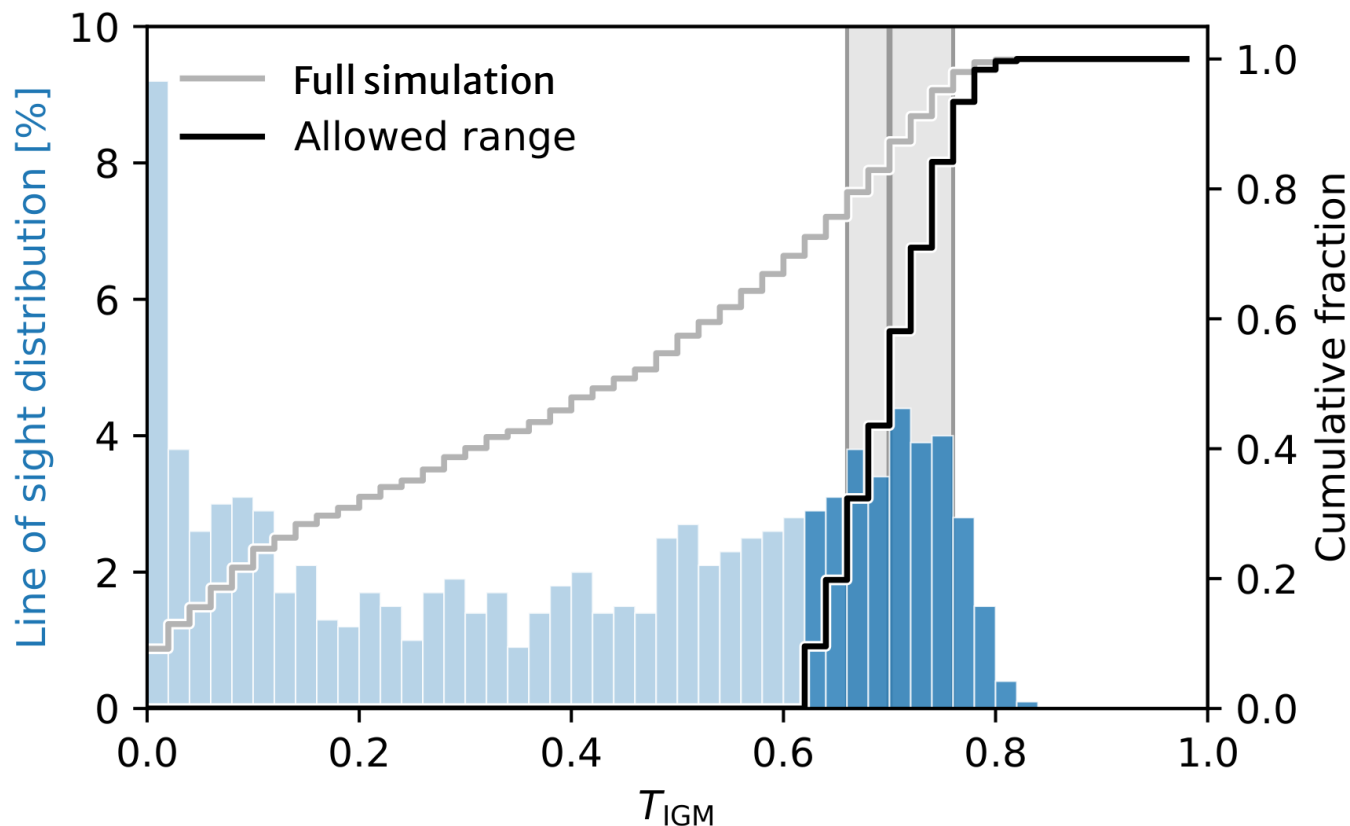


V. Ramachandran+, 2021

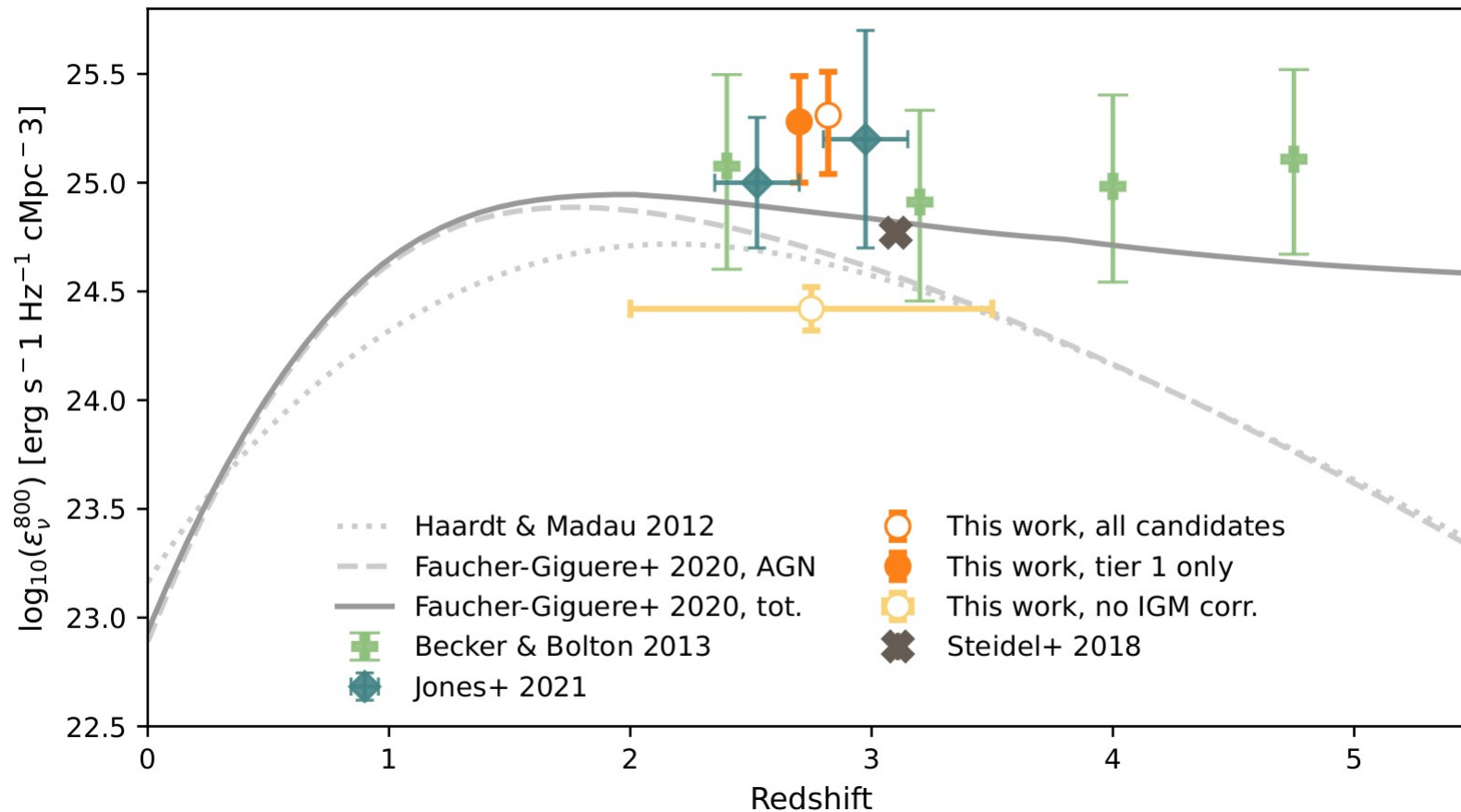
Can we really expect the $\lambda_{\text{rest}}=900 \text{ \AA}$ photons
to experience the **same dust** or gas
column as the averaged $\lambda_{\text{rest}}=1500 \text{ \AA}$ photons?

...And what does that mean for our understanding
of relative and absolute escape?

Simulated line-of-sight **IGM Transmission** - might also skew our interpretation



We found a comoving **volume emissivity** larger than, but consistent with, literature values





Summary:

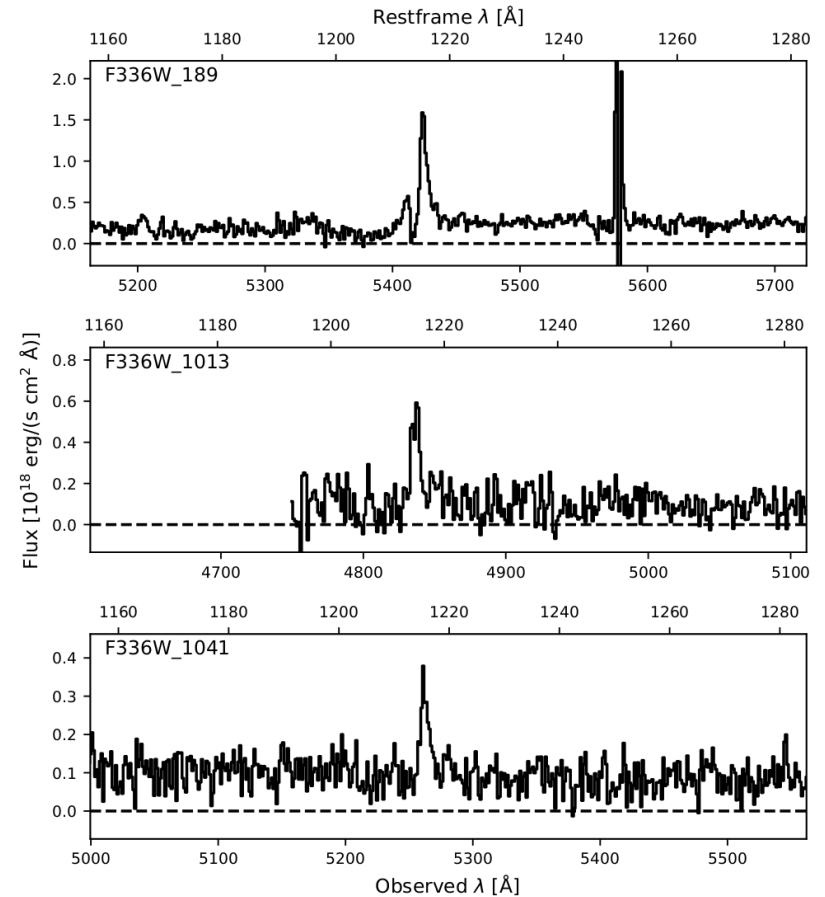
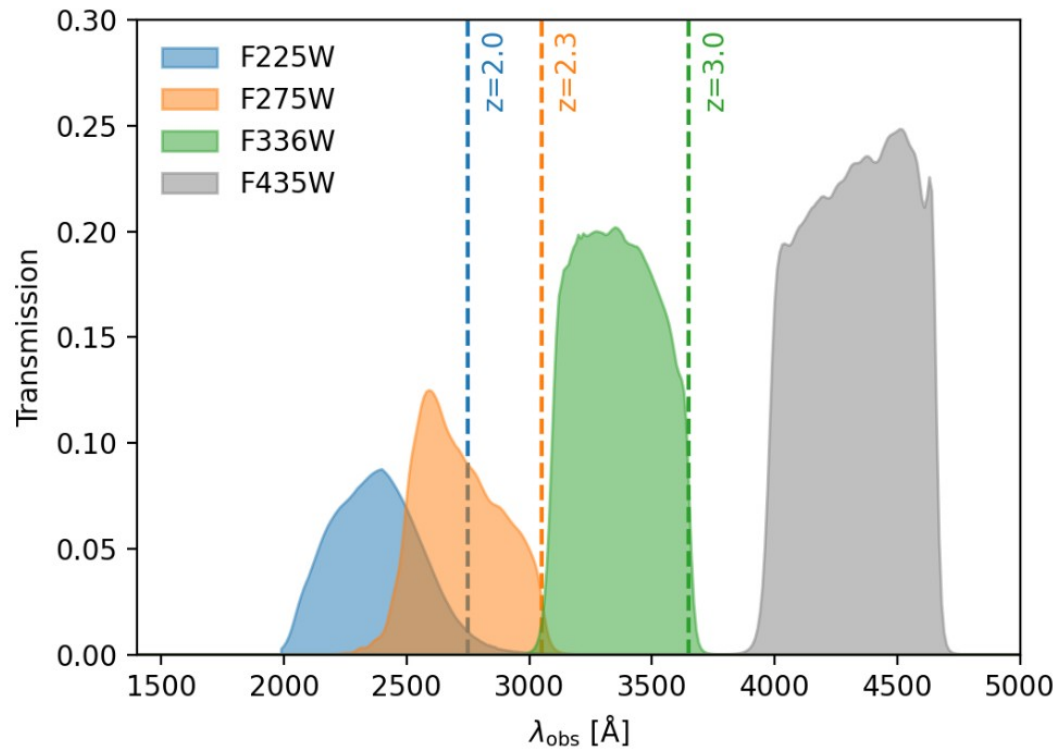
- We found **7 candidates**. One was found in other works. We did not find other candidates reported there.
- We found very **high $f_{\text{esc,rel}}$** , but all $f_{\text{esc,abs}}$ were $\lesssim 100\%$. We argue that this is due to uneven dust geometry and a small number of stars dominating the LyC emission.
- Only three candidates had detected **Ly α emission!**
- Only leaker with **double-peaked** Ly α has $v_{\text{sep}} \approx 620$ km/s
- One candidate clearly emits LyC from its **outskirts**
- 80% of the inferred volume emissivity is contributed by **one LAE galaxy**, F336W–189 – but highly sensitive to T_{IGM} .

Thank you!



Supplementary slides

Filter overview and Ly α emission



Properties of LCE candidates from this work



Table 4: Best fit physical properties of leaker candidates

Filter Object	$\log_{10}(M_{\star})$ $M_{\odot}$	Age Gyr	τ_{del} Gyr	A_V mag	z	$\text{EW}_{\text{Ly}\alpha, \text{obs}}$ $\text{\AA}$
F275W-314	$9.34^{+0.01}_{-0.01}$	$3.20^{+0.03}_{-0.07}$	$6.58^{+2.25}_{-2.35}$	$0.24^{+0.02}_{-0.03}$	2.00*	—
F275W-2055	$9.10^{+0.27}_{-0.34}$	$1.07^{+0.90}_{-0.60}$	$5.69^{+2.96}_{-3.04}$	$0.43^{+0.09}_{-0.12}$	$2.33^{+0.29}_{-0.29}$	—
F336W-189	$9.58^{+0.01}_{-0.03}$	$0.22^{+0.01}_{-0.02}$	$0.13^{+0.01}_{-0.02}$	$0.56^{+0.01}_{-0.01}$	3.46*	127 ± 7
F336W-554	$8.82^{+0.04}_{-0.04}$	$0.15^{+0.02}_{-0.02}$	$3.71^{+4.18}_{-2.99}$	$0.82^{+0.02}_{-0.02}$	2.87*	—
F336W-606	$8.51^{+0.03}_{-0.04}$	$0.10^{+0.02}_{-0.02}$	$0.04^{+0.02}_{-0.01}$	$0.00^{+0.02}_{-0.02}$	$2.73^{+0.01}_{-0.01}$	—
F336W-1013[†]	$8.50^{+0.01}_{-0.01}$	$0.08^{+0.01}_{-0.01}$	$4.57^{+3.73}_{-3.32}$	$0.98^{+0.01}_{-0.01}$	2.98*	118 ± 30
F336W-1041	$9.46^{+0.03}_{-0.03}$	$1.79^{+0.06}_{-0.12}$	$0.48^{+0.03}_{-0.03}$	$0.25^{+0.06}_{-0.04}$	3.33*	210 ± 50

Table 5: Escape fractions and IGM correction

Object	$f_{\text{esc}}^{\text{abs}} \times T_{\text{IGM}}^a$	$f_{\text{esc}}^{\text{rel}} \times T_{\text{IGM}}^b$	T_{IGM}	$f_{\text{esc}}^{\text{rel}}$	$f_{\text{esc}}^{\text{abs}c}$
F275W-314	$51 \pm 33\%$	57%	$72 \pm 9\%$	$82 \pm 12\%$	$77 \pm 9\%$
F275W-2055	$57 \pm 19\%$	105%	$66 \pm 8\%$	$153 \pm 14\%$	$76 \pm 9\%$
F336W-189	$3 \pm 0.8\%$	23%	$7 \pm 5\%$	$325 \pm 200\%$	$36 \pm 22\%$
F336W-554	$28 \pm 7\%$	203%	$49 \pm 11\%$	$441 \pm 110\%$	$67 \pm 15\%$
F336W-606	$40 \pm 17\%$	67%	$42 \pm 3\%$	$146 \pm 12\%$	$91 \pm 6\%$
F336W-1013	$4 \pm 0.6\%$	103%	$24 \pm 15\%$	$442 \pm 295\%$	$15 \pm 10\%$
F336W-1041	$108 \pm 30\%$	397%	$100 \pm -\%$	$397 \pm -\%$	$100 \pm -\%$

Properties of LCE candidates from this work

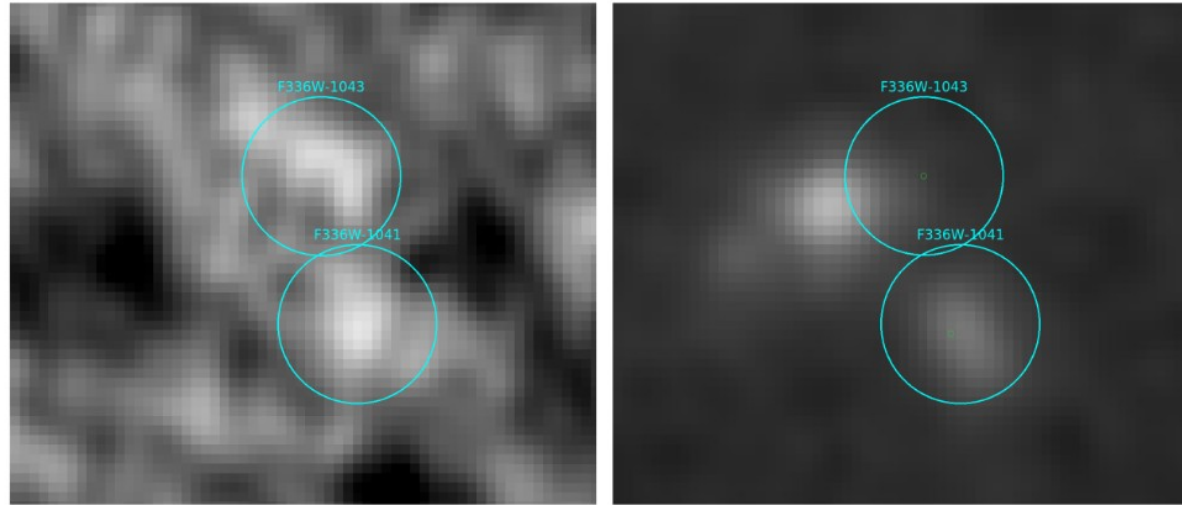


Fig. E.1: Cutouts of F336W-1041 (lower circle) and F336W-1043 (upper circle) in the filters F336W (left panel) and F775W (right panel). The circles are of arbitrary size. The upper feature in F336W is visibly offset from the upper feature in F775W, which is true for the rest of optical and IR data. In contrast, the spatial coincidence of the lower feature in the two filters is clear.